

Communication & Network Fundamentals – (CIS 315)



Report on course project

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Abstract

This project involves designing and implementing a comprehensive network for an organization with multiple branches, utilizing advanced routers, switches, and access point configurations. The network employs three routers (Cisco 1841) and three switches (Cisco 2960-24PT) to establish interconnected sub-networks, ensuring reliable communication and efficient resource sharing. Each router is configured with specific serial and Fast Ethernet interfaces, enabling seamless inter-branch connectivity and Internet access. The switches facilitate local connections among PCs, laptops, printers, servers, and wireless access points within each sub-network.

The topology integrates physical and logical components, creating a structured yet scalable framework. The logical topology connects four sub-networks through serial links, ensuring secure data transmission. VLANs segment the network based on user roles, such as students, employees, and administrators, to enhance security and optimize performance. The physical topology connects devices in a star configuration for robust and efficient communication within each branch.

Wireless access points expand connectivity options for mobile users, while routers handle inter-branch routing and external Internet communication. The network design was validated using simulation tools, confirming its scalability, reliability, and adaptability for future expansion. This project demonstrates a structured approach to building a flexible, secure communication network for multi-branch organizations.

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Introduction

Background

Designing an integrated network for multiple branches requires the use of advanced devices to ensure efficient communication and resource management. Networks control the flow of data within an organization, directing data packets from the originator to the intended receiver.

This project utilizes Cisco 1841 routers that operate in the network layer of the OSI model, which provide connectivity between different branches through serial and Fast Ethernet interfaces, enabling reliable data exchange and Internet access between departments and connecting it to the cloud which has all the information for the company that they need. It has a WAN port for connecting a modem and four LAN ports for connecting devices. The router enables network address translation (NAT) for numerous devices to share a single public IP address

The network also incorporates Cisco 2960-24PT switches that operate in the second layer of the OSI model, which connect devices within each branch, including PCs, printers, and servers, with full-duplex support for efficient communication. To enhance wireless connectivity, wireless access points are included, offering flexibility for mobile users.

A wireless router is a device that connects wireless devices to a network by combining routing and wireless access point operations. It supports 802.11n Wi-Fi protocols and has WAN and LAN interfaces.

Peripherals in the network include servers for services like file sharing, PCs, laptops for user tasks, and printers for hard copies.

Objectives

1. Explain how you create, configure, and verify a small network of directly connected LANs.
2. Explain how to configure routers.

3. Configure the IP addresses, subnet mask, and host addresses for each subnet.
4. Design and configuration.
5. How to design, configure, and verify VOIP over a very small network.

Network and End Devices used

Device	Type	Quantity
PC	PT	5
IP phone	7968	2
LAPTOP	PT	5
Printer	PT	2
Switch	2960-24PT	3
Router	1841	2
Router	2911	1
Access Point	PT	2
Server	PT	2
Modem	DSL-PT	1
Cloud	PT	1
Cable 1	Copper Cross-over	16

Cable 2	Phone	1
Cable 3	Serial DCE	2

Table 1: Devices

Network design

Logical topology

Topology of network used:

Hybrid topology, the Figure depicts a hybrid topology network with Star Topology subnets that contain PCs, laptops, printers, and servers connected to switches and access points. And Tree/ Hierarchical Topology by the routers to branch different subnets with mesh elements between the routers that have redundant connections. Those elements contain leveraging strengths of multiple designs for flexibility, scalability, and fault tolerance. However, it also has its drawbacks, like complex design and setup, dependency on multiple protocols, management overhead, high cost, and difficult troubleshooting.

Key characteristics of hybrid topology include a combination of multiple topologies, scalability, flexibility, customizability, fault tolerance, centered or decentralized control, heterogeneous communication its ablest communication between devices and systems that has different protocols or topology, performance optimization it performs based on the segment best suited for.

Hybrid topologies are well-suited for large organizations and multi-site operations to ensure effective connectivity between sites and for high availability and fault tolerance requirements.

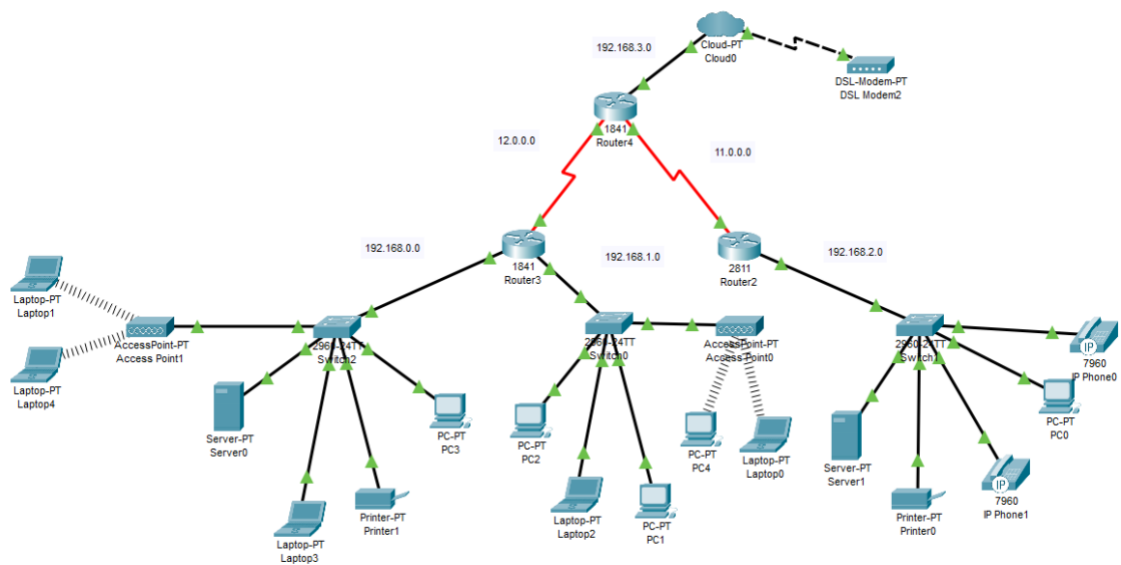


Figure 1 : The Network

Physical Topology



Figure 2: Physical Topology

Addressing table

Device	Interface	IP address	Subnet Mask	Default Gateway
DSL Modem2	N/A	N/A	N/A	N/A
Cloud0	Ethernet9	N/A	N/A	N/A
Router4	FastEthernet0/0	192.168.3.10	255.255.255.0	N/A
Router4	Serial0/1/0	12.0.0.1	255.255.255.252	N/A
Router4	Serial0/1/1	11.0.0.1	255.255.255.252	N/A
Router2	FastEthernet0/0	192.168.2.1	255.255.255.0	N/A
Router2	Serial0/3/0	11.0.0.2	255.255.255.252	N/A
Router3	FastEthernet0/0	192.168.1.1	255.255.255.0	N/A
Router3	FastEthernet0/1	192.168.0.1	255.255.255.0	N/A
Router3	Serial0/1/0	12.0.0.2	255.255.255.252	N/A
Switch1	FastEthernet0/1	N/A	N/A	N/A
Switch1	FastEthernet0/2	N/A	N/A	N/A
Switch1	FastEthernet0/3	N/A	N/A	N/A
Switch1	FastEthernet0/4	N/A	N/A	N/A
Switch1	FastEthernet0/5	N/A	N/A	N/A
Switch1	FastEthernet0/6	N/A	N/A	N/A
Switch1	FastEthernet0/7	N/A	N/A	N/A
Switch0	FastEthernet0/1	N/A	N/A	N/A
Switch0	FastEthernet0/2	N/A	N/A	N/A
Switch0	FastEthernet0/3	N/A	N/A	N/A
Switch0	FastEthernet0/4	N/A	N/A	N/A
Switch0	FastEthernet0/5	N/A	N/A	N/A

Switch2	FastEthernet0/1	N/A	N/A	N/A
Switch2	FastEthernet0/2	N/A	N/A	N/A
Switch2	FastEthernet0/3	N/A	N/A	N/A
Switch2	FastEthernet0/5	N/A	N/A	N/A
Switch2	FastEthernet0/4	N/A	N/A	N/A
Access Point 0	Port 1	N/A	N/A	N/A
Access Point 1	Port1	N/A	N/A	N/A
Laptop0	Wireless0	192.168.1.5	255.255.255.0	192.168.1.1
Laptop1	Wireless0	192.168.0.7	255.255.255.0	192.168.0.1
Laptop2	FastEthernet0	192.168.1.2	255.255.255.0	192.168.1.1
Laptop3	FastEthernet0	192.168.0.3	255.255.255.0	192.168.0.1
Laptop4	Wireless0	192.168.0.9	255.255.255.0	192.168.0.1
PC0	FastEthernet0	192.168.2.4	255.255.255.0	192.168.2.1
PC4	Wireless0	192.168.1.6	255.255.255.0	192.168.1.1
PC1	FastEthernet0	192.168.1.4	255.255.255.0	192.168.1.1
PC2	FastEthernet0	192.168.1.3	255.255.255.0	192.168.1.1
PC3	FastEthernet0	192.168.0.6	255.255.255.0	192.168.0.1
Server0	FastEthernet0	192.168.0.2	255.255.255.0	192.168.0.1
Server1	FastEthernet0	192.168.2.2	255.255.255.0	192.168.2.1
Printer0	FastEthernet0	192.168.2.3	255.255.255.0	192.168.2.1
Printer1	FastEthernet0	192.168.0.4	255.255.255.0	192.168.0.1
IP Phone0	FastEthernet0/6	192.168.2.5	255.255.255.0	192.168.2.1
IP Phone1	FastEthernet0/4	192.168.2.6	255.255.255.0	192.168.2.1

Table 2 : The Addressing Table

creating, configuring, and verifying a small network of directly connected LANs explaining.

There are 4 networks in this project: the first network has switch 2 directly connected to Server0, Laptop3, Printer1, PC3, and Access point1 which is wireless connected to two devices Laptop1 and Laptop4. The second network has switch0 directly connected to PC2, Laptop2, PC1, and Access point0 which is wireless connected to two devices PC4 and Laptop0. The third network has switch1 directly connected to Server1, PC0, and Printer1. The fourth network has Cloud0, which connects to DLS Modom2.

Configuring The IP addresses, Subnet Mask, and Host Addresses for Each Subnet

We manually did the Configuration of IP addresses for the 3 networks (SW0, SW1, SW2). Nevertheless, The Host Addresses have been manually added for each network device. Moreover, the Subnet Mask for all the Host Addresses devices was set to (255.255.255.0) and all the addresses are mentioned in Table 2.

How Routers Configured

There are two ways to configure the router either through the config tab or the CLI, we used the config tab. In router4 has serial0/1/0 is 12.0.0.1 which is connected to hop 12.0.0.2 and serial0/1/1 is 11.0.0.1 which is connected to hop 11.0.0.2, router3 has serial0/1/0 12.0.0.2 which is connected to hop

12.0.0.1. And in router0 has serial0/1/0 11.0.0.2 which is connected to hop 11.0.0.1.

Connection Test

As you can see from the figure below this was a simple test using the message feature in the Cisco packet tracer. We have sent a message from a couple of random devices to other devices across the network, as you see the messages were successful.

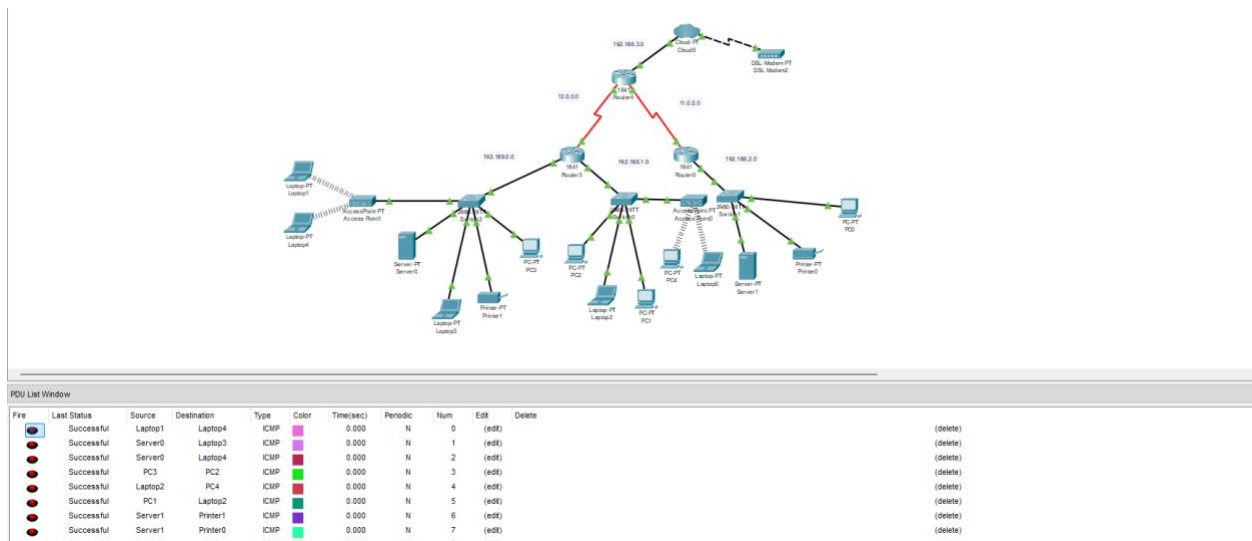


Figure 3: Connection Test



Double Click to watch

Conclusion

In conclusion, this network has a solid and effective design to meet the needs of the office. By using a clear hybrid structure and rationally distributing devices, the project's objectives of connecting the three departments efficiently and safely were achieved. This network has improved communication between employees, facilitated data exchange, and increased productivity. They also provide the basis for future expansions, where new hardware or subnets can be added without the need for a complete network redesign

This network relies on various modern technologies, including routers, switches, and wireless access points. Appropriate routing protocols have been used to ensure reliable data delivery. Strict security measures have also been implemented to protect the network from external threats. By integrating these technologies, a flexible and expandible network capable of supporting a wide range of applications has been built.